

Roll Number: _____

Thapar Institute of Engineering & Technology
Department of Electronics and Communication Engineering

M.E. (II SEM): VLSI

Course Code: PVL333

Course Name: System On Chip

March 14, 2026, 3:00 PM

Time: 02 Hrs.

Max. Marks: 30.

Name of Faculty: Dr. Robin Singla

Note: Attempt all questions & assume missing data, if any, suitably

Q. No.	Questions	Marks	CLO	BL
Q.1	Explain the following in brief. (a) Mention deep submicron effects (b) Different components of SOC system (c) Hardware vs Software: Programmability vs Performance (d) H/W – S/W co-design (e) Top-down SOC design flow	(10)	CLO1	L1 & L2
Q.2	What is the role of on-chip interconnects? Explain the fundamental difference between BUS and NOC. Use diagrams wherever required.	(3+2)	CLO2	L1 & L2
Q.3	Explain the NOC architecture in detail.	(5)	CLO2	L2 & L3
Q.4	Explain in detail the five chip basics.	(5)	CLO1	L2 & L3
Q.5	What are the SOC design metrics? Which different techniques can be used to improve a specific design metric?	(5)	CLO4	L2 & L3
